UNIVERSITÀ DEGLI STUDI DELL'AQUILA

Dipartimento di Ingegneria e Scienze dell'Informazione e Matematica

Tesi di Laurea Magistrale in Ingegneria Matematica

Predicting post-operative outcomes related to patients who underwent haemorrhoids surgery in an Italian hospital.

Supervisor
Dr. Andrea Manno

Co-Supervisor
Prof. Fabrizio Rossi

Author:
Abdoulie J Bah
287149

Anno Accademico 2023-2024

Some quote...

Abstract

Post-operative complications following haemorrhoids surgery pose significant challenges to patient recovery and healthcare management. This project aims to develop a predictive model for identifying patients at risk of complications post-haemorrhoids surgery in an Italian hospital setting, with a focus on the binary-class of Complication and Relapses. Leveraging a dataset comprising clinical, and procedural variables collected from haemorrhoids surgery patients, advanced statistical techniques and machine learning algorithms will be employed to develop a robust predictive model. The model's accuracy in classifying patients into high-risk and low-risk categories will be evaluated, with the ultimate goal of enhancing patient outcomes, refining clinical decision-making, and optimizing resource allocation. Insights gained from this study have the potential to help the management of haemorrhoids surgery patients, facilitating more personalized and proactive care delivery in the realm of colorectal surgery.

Keywords

Hemorrhoids surgery, Post-operative outcomes, Complications prediction, Binary classification, Italian hospital, Predictive modeling, Healthcare management

Contents

List of Figures **iv**

List of Tables **v**

Nomenclature **vii**

1 Introduction **1**

2 Literature Review **3**

3 The Problem and the Dataset **6**
3.1 The Problem Description 6
3.2 Dataset Description 7
3.3 Dataset Analysis 8

4 Machine learning Methods **12**
4.1 Overview of Machine Learning Algorithms 12
4.2 The Adopted Machine Learning Methods 13

5 Data Pre-processing and Experimental Settings **18**
5.1 Preprocessing 18
5.2 Experimental Settings 19
5.3 Data Imbalance Handling 19

6 Results **21**
6.1 Preliminary Results 21
6.2 Complication Binary 21
6.3 Relapses Binary 31

7 Conclusions **41**

References **43**

List of Figures

2.1 Flowchart of ML Integration in Clinical Workflows 4

3.1 Distribution of Key Numerical Variables 9
3.2 Class Distribution Across Target Variables 9
3.3 Correlation Heatmap of Features 10
3.4 Outlier Analysis of Numerical Features 10
3.5 Missing Values Across Variables 11

List of Tables

3.1 Italian Terms and Their English Meanings 7

6.1 Complication Binary with 40 percent re-balance class 24
6.2 Complication Binary with 50 percent re-balance class 25
6.3 Complication Binary with 60 percent re-balance class 26
6.4 Complication Binary with 65 percent re-balance class 27
6.5 Complication Binary with 75 percent re-balance class 28
6.6 Complication Binary with 85 percent re-balance class 29
6.7 Complication Binary with 100 percent re-balance class 30
6.8 Relapses Binary with 40 percent re-balance class 34
6.9 Relapses Binary with 50 percent re-balance class 35
6.10 Relapses Binary with 60 percent re-balance class 36
6.11 Relapses Binary with 65 percent re-balance class 37
6.12 Relapses Binary with 75 percent re-balance class 38
6.13 Relapses Binary with 85 percent re-balance class 39
6.14 Relapses Binary with 100 percent re-balance class 40

Nomenclature

M_∞	Free Stream Mach number
M_∞	Free Stream Mach number

Chapter 1

Introduction

Hemorrhoids are a widespread medical condition, with severe cases often necessitating surgical intervention for effective symptom management. Despite advances in surgical techniques, post-operative complications continue to be a significant concern, adversely affecting patient recovery [9]. In an effort to reduce these complications, a variety of surgical approaches have been proposed [31]. This thesis focuses on predicting post-operative outcomes for patients who underwent hemorrhoid surgery at an Italian hospital, using machine learning (ML) methods to better understand and improve recovery processes [18].

Post-operative complications and relapses are critical factors that significantly impact patient outcomes and overall healthcare quality [21]. Anticipating and predicting these outcomes is essential for enhancing clinical decision-making, optimizing patient management, and effectively allocating healthcare resources [23]. Given the high volume of hemorrhoid surgeries performed, the variation in outcomes, and the associated risks, there is a pressing need for data-driven approaches to refine predictions and mitigate potential risks [25]. This thesis aims to employ advanced ML techniques to predict post-operative complications and relapses, ultimately improving patient care by identifying potential issues early [27].

In recent years, ML has emerged as a transformative tool in healthcare, especially in post-operative care [3]. By analyzing large, complex datasets, ML models can identify patterns and correlations that might be missed by traditional clinical methods [26]. These models have proven effective in forecasting post-operative risks by extracting actionable insights from structured data [7]. Integrating ML into existing frameworks, such as the Modified Early Warning Scoring (MEWS) system, can help healthcare providers proactively manage potential complications [14]. By prioritizing the collection of granular and standardized data, ML can facilitate targeted interventions that reduce the likelihood of complications and optimize overall surgical outcomes [2].

This thesis aims to extend the potential of ML in predicting post-operative complications and relapses for hemorrhoid surgery patients at an Italian hospital [13]. By addressing issues such as data imbalance and striving to extract meaningful insights, this study seeks to improve clinical outcomes, reduce healthcare costs, and contribute to more efficient patient management strategies [19].

The task of predicting post-operative complications and relapses presents several challenges:

1. **Data Imbalance:** A skewed distribution of outcomes—where one class is much larger than another—can lead to biased models that perform inadequately for minority classes.

2. **Complex Target Variables:** Complications and relapses manifest as both binary and multi-class targets, necessitating distinct modeling approaches for each case.
3. **Data Quality:** The presence of null values and outliers within the dataset complicates the analysis, requiring meticulous preprocessing.

Building upon existing research that underscores the necessity for standardized and validated ML approaches in perioperative care, this thesis aims to face these challenges. By employing rebalancing techniques such as SMOTE and evaluating various machine learning models, this study aspires to uncover robust solutions that enhance patient outcomes[29].

The thesis involves segmenting the dataset into four subsets: complication binary, relapses binary, complication multi-class, and relapses multi-class. The findings may be influenced by the size, quality, and specificity of the dataset, which could affect the developed models' applicability to other populations or healthcare settings. Furthermore, the complexity of the models might be shaped by the computational resources available.

The thesis is structured as follows:

1. **Introduction:** An overview of the research background, motivation, problem statement, and scope, laying the groundwork for improved predictions of post-operative outcomes via machine learning.
2. **Literature Review:** A review of existing studies on surgical outcome prediction and machine learning applications in healthcare, highlighting the gaps this thesis aims to fill.
3. **The Problem and the Dataset:** A detailed description of the dataset utilized in the study, addressing its structure, challenges such as data imbalance, and the rationale behind the dataset subdivision.
4. **Machine Learning Methods:** An introduction to the machine learning models employed, including their relevance to the research problem, and a discussion of evaluation metrics (e.g., balanced accuracy, precision, recall) for assessing performance.
5. **Data Preprocessing and Experimental Settings:** Coverage of preprocessing steps (e.g., null value management, outlier handling, data normalization) and the experimental setup, including training-test splits and hyperparameter tuning.
6. **Results:** Presentation of findings across the datasets, comparing model performances and incorporating detailed analyses using evaluation metrics and visualizations.
7. **Conclusions:** A summary of research contributions, a discussion on limitations, and suggestions for future work, reflecting on the implications of the findings for clinical practice.

Chapter 2

Literature Review

The application of machine learning (ML) in healthcare has seen a rapid expansion, revolutionizing the analysis and utilization of medical data. ML techniques have enabled the analysis of complex datasets, such as genetic data, physiological imaging, and electronic health records (EHRs)[32]. Machine learning involves designing and developing systems capable of learning from data and making predictions without explicit programming. It is especially valuable in contexts where data is generated more rapidly than humans can process. In intensive care settings, where managing care is complex and vast amounts of data are produced, machine learning presents an ideal opportunity to aid clinical decision-making by analyzing this data effectively[22]. These algorithms excel at identifying patterns and relationships in data, often surpassing human analytical capabilities [23]. By examining historical medical data, ML can predict patient outcomes, empowering healthcare providers to intervene early and mitigate potential complications. These capabilities have been demonstrated in areas such as pandemic forecasting, personalized treatment strategies, and chronic disease management [2].

Research has further shown the integration of ML into clinical workflows to be effective in identifying high-risk patients, optimizing healthcare resources, and improving treatment outcomes. For instance, predictive analytics have been applied to early detection of sepsis, prediction of hospital readmissions, and optimizing surgical schedules[15]. We analyzed the predictive performance of various models, including logistic regression, generalized additive models, naïve Bayes, and support vector machines, for identifying postoperative sepsis and acute kidney injury. The study also examined the influence of feature reduction techniques on model accuracy. Performance was measured using metrics such as the area under the receiver operating characteristic (ROC) curve, accuracy, and positive predictive value. A 70/30 cross-validation approach was employed, with 70 percent of the data allocated for model training and the remaining 30 percent for validation[6]. Despite these advancements, several challenges persist, including the interpretability of complex models, the presence of imbalanced datasets, and the need for strict data privacy compliance [19].

Within surgical outcome prediction, ML has been successfully used to predict complications, assess risks, and optimize post-surgical care. For example, studies employing decision trees, support vector machines (SVMs), and ensemble methods have demonstrated their ability to predict surgical site infections, identify patients at high risk for reoperation, and estimate recovery times[1]. These findings highlight the utility of ML in tailoring post-operative care and enhancing clinical decision making[11].

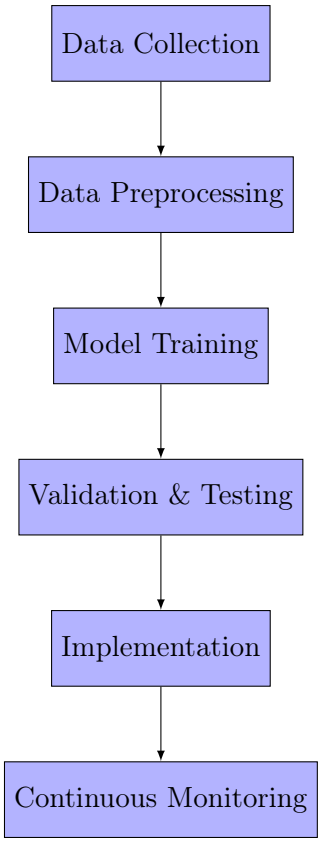


Figure 2.1: Flowchart of ML Integration in Clinical Workflows

Specifically for hemorrhoid surgery, existing literature primarily focuses on clinical trials and comparisons of surgical techniques rather than leveraging ML for predictive purposes. Studies have evaluated the effectiveness of surgical approaches such as stapled hemorrhoidopexy and conventional hemorrhoidectomy, noting varying rates of complications and relapses. However, there is limited research on how data-driven models can predict these outcomes preemptively, creating a significant gap in the literature [34].

This thesis seeks to bridge this gap by applying ML models to predict post-operative complications and relapses in patients undergoing hemorrhoid surgery. This study, based on data from an Italian hospital, aims to demonstrate how machine learning can enhance the understanding of individual risk profiles and inform tailored interventions. By addressing challenges like class imbalance, the research seeks to ensure robust and actionable results. It also explores advanced techniques, such as grid search and data re-balancing, to improve model reliability and clinical applicability.

Many healthcare datasets suffer from class imbalance, which impacts the reliability and fairness of predictive models. This thesis addresses this gap by employing advanced techniques such as data re-balancing and grid search optimization[28]. While machine learning has shown potential in intensive care, its application to support clinical decision-making in specific conditions like post-operative sepsis and acute kidney injury remains under explored[10]. This thesis seeks to bridge this gap by focusing on these critical areas. The lack of attention to effective feature selection and interpretability in predictive models limits their clinical applicability[30]. This work explores feature reduction methods to enhance model transparency and reliability. Existing studies often employ limited evaluation metrics or procedures[16]. By incorporating robust metrics, such as the area

under the ROC curve, accuracy, and positive predictive value, and using rigorous cross-validation techniques, this thesis ensures comprehensive performance assessment[16].

Chapter 3

The Problem and the Dataset

3.1 The Problem Description

The application of machine learning (ML) in healthcare offers significant potential for enhancing diagnostics, treatment strategies, and overall patient outcomes. However, applying ML to medical data is fraught with unique challenges stemming from the complexity, diversity, and inconsistencies inherent in such datasets. This thesis seeks to address these challenges in the context of predicting post-operative outcomes—specifically complications and relapses—following hemorrhoid surgery. The ultimate goal is to create robust predictive models that can aid in clinical decision-making, thus improving recovery processes and resource allocation.

The dataset used in this research has two primary target variables: **complications** and **relapses**. Each of these target variables is further divided into binary and multi-class variants. This division reflects the nuanced nature of medical outcomes and allows for more precise modeling of different clinical scenarios. Consequently, the dataset is split into four distinct subsets: *complication binary*, *complication multi-class*, *relapse binary*, and *relapse multi-class*. Each subset is analyzed independently to ensure that the unique characteristics and challenges associated with each target type are adequately addressed.

A significant issue across all four subsets is the problem of **class imbalance**. In medical datasets, certain outcomes—such as severe complications or relapses—tend to occur less frequently than others, resulting in a skewed distribution of classes. This imbalance poses a challenge for ML models, which often perform poorly on minority classes due to their bias toward the majority class. To tackle this issue, this research employs the Synthetic Minority Oversampling Technique (SMOTE), which generates synthetic examples for underrepresented classes. This approach helps rebalance the data and improves the models' ability to predict rare but critical outcomes.

Another challenge is the presence of **missing data**. Medical records often have incomplete information due to a variety of factors, such as gaps in data entry, patient non-compliance, or unavailability of certain diagnostic results. Missing data can reduce the reliability of ML models and skew results if not handled properly. In this study, missing values are addressed during the preprocessing phase using imputation techniques, ensuring that the dataset remains as complete and consistent as possible.

The issue of **outliers** is also prevalent in medical datasets. Outliers—extreme data points that deviate significantly from the rest of the dataset—can arise due to errors in measurement, recording, or

rare patient conditions. These outliers can distort statistical analyses and negatively impact model training, especially in sensitive applications like healthcare. To mitigate this, robust methods for detecting and managing outliers are applied, ensuring that they do not disproportionately influence the results while retaining clinically relevant information.
In addition, the **heterogeneity of features** within the dataset necessitates normalization. Medical datasets often contain a mix of numerical measurements, categorical variables, and binary indicators. The varying scales of these features can hinder model convergence and performance if left unaddressed. Data normalization techniques are applied to standardize the scales of numerical features, enabling the ML models to process the data more effectively.
Overall, this thesis addresses the intertwined challenges of handling imbalanced datasets, missing values, outliers, and the need for data normalization. By dividing the dataset into four subsets and employing targeted preprocessing techniques, this research creates a robust foundation for developing and evaluating predictive models. These efforts aim to improve clinical outcomes by providing accurate predictions of post-operative complications and relapses, facilitating timely interventions, and optimizing patient care.

3.2 Dataset Description

The dataset utilized in this study is sourced from an Italian hospital and contains 17 variables that describe patient demographics, clinical observations, preoperative assessments, and postoperative outcomes. The variables are detailed in Table 3.1, with translations provided for clarity. This dataset forms the foundation for analyzing and predicting postoperative complications (*complicanze*) and relapses (*recidive*), which are the primary focus of this study.

Table 3.1: Italian Terms and Their English Meanings

Italian Term	**English Meaning**
ETA'	Age
SESSO	Gender
PROLASSO_EMORR	Hemorrhoid Prolapse
tipo_interv	Type of Intervention
Ligasure	Ligasure (Surgical Tool)
Focus	Focus (Surgical Technique)
Diatermica	Diathermic
anestesia	Anesthesia
SANGUINAMENTO_npre	Bleeding (Preoperative)
RIDUZIONEMANUALE_npre	Manual Reduction (Preoperative)
DOLORE-FASTIDIOpre	Pain-Discomfort (Preoperative)
scoretotale_pre	Preoperative Total Score
COMPLICANZE	Complications (Binary)
RECIDIVE	Relapses (Binary)

The dataset includes two primary target variables, *complicanze* (complications) and *recidive* (relapses), each of which is divided into binary and multi-class formats. Consequently, the dataset is split into four distinct subsets for independent analysis: binary *complicanze*, multi-class *complicanze*, binary *recidive*, and multi-class *recidive*. Each subset excludes columns related to the other target formats to ensure a focused and unbiased analysis.

Preprocessing the dataset involved addressing several challenges inherent to medical data. Missing values were handled using imputation techniques to ensure data completeness without losing critical information. Outliers were identified and managed carefully to retain clinically relevant variations while minimizing their potential impact on model training. Data normalization was applied to scale features consistently, facilitating improved model performance.
A significant issue identified in the dataset was class imbalance in all four target subsets. For instance, certain classes of complications and relapses were underrepresented, posing challenges for training machine learning models. To address this, the Synthetic Minority Oversampling Technique (SMOTE) was employed to generate synthetic samples of minority classes, thereby improving the balance and reliability of the data.
Overall, the dataset provides a rich and complex source of information for exploring the relationship between clinical features and postoperative outcomes. The preprocessing and careful subdivision into target-specific datasets lay the groundwork for robust machine learning analysis aimed at improving predictive accuracy and contributing to enhanced patient care.

3.3 Dataset Analysis

This section provides an in-depth analysis of the dataset to gain insights into its structure, distribution, and key characteristics. The analysis includes examining the dataset's variables, identifying patterns and relationships among features, and evaluating the challenges associated with each of the four target variables.
The first step in analyzing the dataset was understanding its structure and distribution. The dataset comprises 17 variables, encompassing patient demographics, preoperative assessments, surgical techniques, and postoperative outcomes. An exploration of the data revealed variations in the distribution of numerical variables, such as age and preoperative scores, as well as the presence of binary indicators for clinical conditions and procedures. The distribution of these variables was visualized using histograms, boxplots, and density plots to identify trends and potential irregularities.

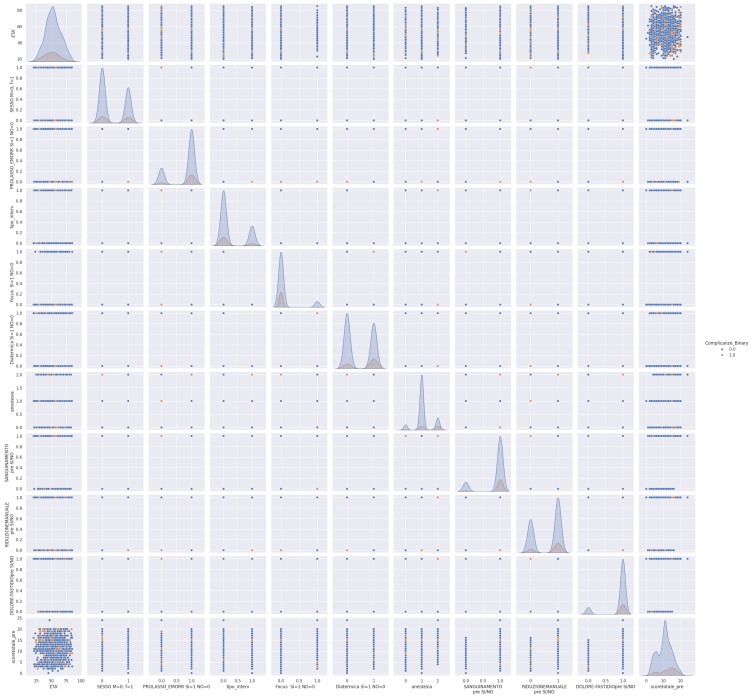


Figure 3.1: Distribution of Key Numerical Variables

A key aspect of the dataset analysis was the evaluation of the target variables. Each of the four subsets—binary *complicanze*, multi-class *complicanze*, binary *recidive*, and multi-class *recidive*—exhibited unique characteristics. For instance, the binary *complicanze* and *recidive* subsets showed class imbalance, with one class significantly underrepresented. Similarly, the multi-class targets revealed skewed distributions, where higher-order complication or relapse categories were rare. To address these imbalances, class distributions were analyzed quantitatively using metrics such as class frequency ratios, and visualized through bar charts to highlight disparities.

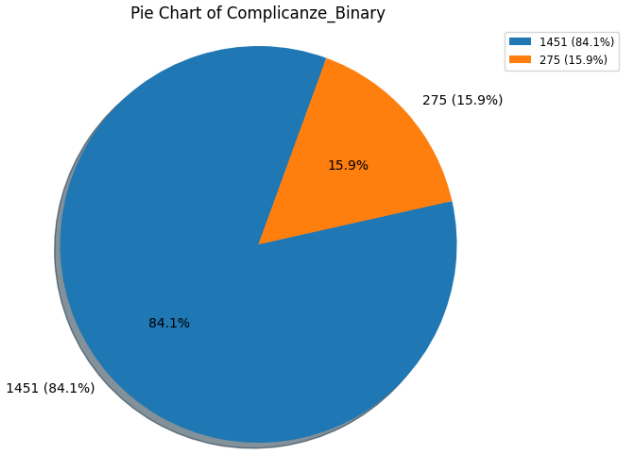


Figure 3.2: Class Distribution Across Target Variables

Correlation analysis was conducted to investigate the relationships between the features and the target variables. Pearson or Spearman correlation coefficients were calculated for numerical variables, while contingency tables and chi-square tests were used for categorical features. This analysis provided insights into which variables were most predictive of complications and relapses, guiding

feature selection in subsequent modeling stages.

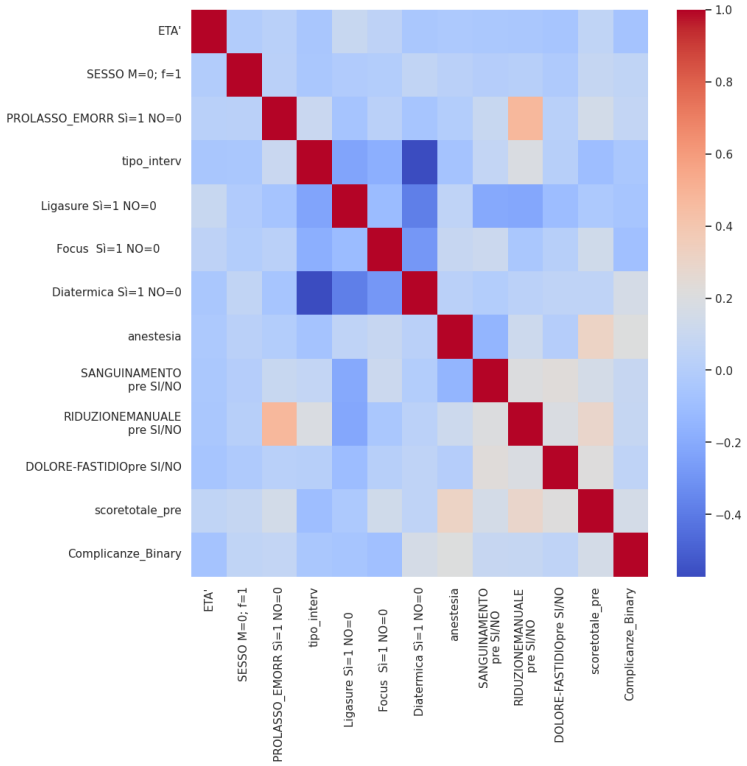


Figure 3.3: Correlation Heatmap of Features

Outlier analysis was performed to identify and manage extreme values that could influence model performance. For instance, preoperative scores and age occasionally displayed extreme deviations. Outliers were visualized using boxplots, and their clinical relevance was assessed to determine whether they represented valid observations or data entry errors.

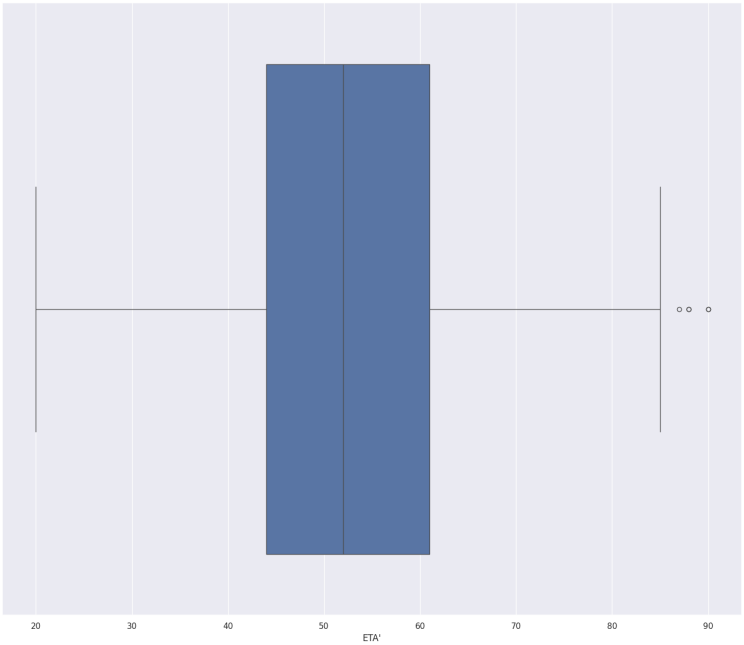


Figure 3.4: Outlier Analysis of Numerical Features

Missing values were another significant aspect of the dataset analysis. Missingness patterns were evaluated to determine whether values were missing at random or exhibited systematic biases. Techniques such as heatmaps and summary tables were employed to visualize missing data. This analysis informed the choice of imputation methods, ensuring that critical information was preserved.

Missing values

```
Binary_Complicanze.isnull().any()
```

	0
ETA'	True
SESSO M=0; f=1	False
PROLASSO_EMORR Sì=1 NO=0	True
tipo_interv	False
Ligasure Sì=1 NO=0	False
Focus Sì=1 NO=0	False
Diatermica Sì=1 NO=0	False
anestesia	False
SANGUINAMENTO\npre SI/NO	True
RIDUZIONEMANUALE\npre SI/NO	True
DOLORE-FASTIDIOpre SI/NO	True
scoretotale_pre	True
COMPLICANZE SI/NO	True

dtype: bool

Figure 3.5: Missing Values Across Variables

Finally, normalization of numerical variables was performed to ensure consistency in feature scaling. Variables such as age and preoperative scores, which spanned different ranges, were scaled using min-max normalization or standardization. This preprocessing step ensured that all features contributed equally to the machine learning models.

Overall, the dataset analysis provided critical insights into the characteristics and challenges of the data. These findings informed the preprocessing steps and the design of predictive models, ensuring a robust approach to addressing the research objectives.

Chapter 4

Machine learning Methods

4.1 Overview of Machine Learning Algorithms

Machine learning (ML) has become an indispensable tool in healthcare, providing the means to harness large, complex datasets for improved clinical decision-making, disease prediction, and operational efficiency. The ability of ML algorithms to identify intricate patterns within data can lead to enhanced diagnostic accuracy, more effective treatment plans, and a higher standard of patient care. This section delves into various machine learning techniques applicable to medical data analysis, examining classical models and advanced ensemble methods, as well as the significance of model evaluation, feature selection, and interpretability.

The ML algorithms employed for medical data analysis range from traditional statistical models to sophisticated machine learning frameworks. Understanding the strengths and weaknesses of each algorithm is essential for choosing the most appropriate method for a given dataset. The following types of algorithms are discussed:

Logistic Regression: A fundamental algorithm often used for binary classification problems, providing clear insights into relationships between features and the target variable. Support Vector Machine (SVM): An effective technique for both linear and non-linear classification problems, known for its ability to handle high-dimensional spaces. Gradient Boosting Classifier: A robust ensemble method that builds models sequentially, focusing on the errors made by previous models to improve overall prediction performance. Naive Bayes: A probabilistic classifier based on Bayes' theorem, suitable for situations where feature independence can be reasonably assumed. Random Forest Classifier: A versatile and powerful ensemble learning method that constructs multiple decision trees to produce a strong, accurate model. K-Nearest Neighbors (KNN): A non-parametric algorithm that classifies data points based on the majority class among their nearest neighbors, known for its simplicity and effectiveness. XGBoost Classifier: A gradient boosting framework that has gained popularity for its speed, scalability, and accuracy in predictive analytics.

Each of these algorithms has unique characteristics that make it suitable for specific types of medical data analysis tasks. The choice of algorithm depends on factors such as the nature of the dataset, the problem at hand, and the need for interpretability. Evaluating these algorithms effectively requires appropriate metrics such as accuracy, precision, recall, F1-score, and balanced accuracy, especially for imbalanced datasets. Additionally, feature selection and model interpretability are critical for ensuring that ML models provide actionable and understandable insights in clinical contexts.

The deployment of ML algorithms in healthcare settings must also consider ethical considerations, data privacy, and the potential for biases in training data. Ensuring that models are both accurate and fair is essential for maintaining trust and equity in healthcare applications. This section sets the stage for a deeper exploration of how these algorithms are applied to predict post-operative outcomes and improve patient care through advanced data-driven approaches.

4.2 The Adopted Machine Learning Methods

This section provides an overview of the machine learning methods adopted for this study. Each method is analyzed in terms of its advantages, disadvantages, mathematical formulations, and the programming packages used for implementation. The models covered include Logistic Regression, Support Vector Machine (SVM), Gradient Boosting Classifier, Naive Bayes, Random Forest, K-Nearest Neighbors (KNN), and XGBoost.

Logistic Regression
The logistic regression model focuses on the concept of odds, defined as the ratio of the probability of an event occurring to its complement. For example, "even odds" occur when the probability of an event, like rolling an even number on a die, is 50%. The model expresses the natural logarithm of these odds as a function of predictors, where for a single predictor X, the model takes the form:

$$\ln[\text{odds}(Y = 1)] = \beta_0 + \beta_1 X.$$

Logistic regression is simple and interpretable, making it efficient for binary classification tasks, especially with linearly separable data. However, it is limited to linear decision boundaries, struggles with multicollinearity and irrelevant features, and assumes a linear relationship between features and the log-odds of the outcome [20].
The mathematical formulation models the probability p of the positive class as:

$$p = \frac{1}{1 + e^{-\mathbf{X}\beta}}$$

where $\mathbf{X}$ represents the features and β the coefficients. The model optimizes the log-likelihood function:

$$\ell(\beta) = \sum_{i=1}^{N} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

This model is commonly implemented using tools such as Python's `scikit-learn` and `statsmodels` libraries or the `glm()` function in R.

Support Vector Machine (SVM) Support Vector Machines (SVM) are a powerful classification technique that builds an optimal hyperplane to separate data points from two different classes. The primary goal is to maximize the margin between the hyperplane and the nearest data points from each class, known as support vectors. This maximized distance ensures good generalization and enhances the predictive capability of the model. SVMs are particularly suitable for high-

dimensional data and can handle non-linear boundaries using kernel functions, mapping the data to higher-dimensional spaces [17].
SVMs can be computationally demanding for large datasets, and selecting the appropriate kernel and tuning hyperparameters can be complex. Additionally, SVMs may not perform well with noisy or overlapping data classes.
For a linear SVM, the decision function is defined as:

$$f(\mathbf{x}) = \text{sign}(\mathbf{w} \cdot \mathbf{x} + b)$$

where $\mathbf{w}$ is the weight vector, $\mathbf{x}$ is the input feature vector, and b is the bias term. Kernel functions are used to map input data into higher-dimensional spaces when dealing with non-linear data.
SVMs can be implemented using various programming packages. In Python, `scikit-learn` is widely used, while in R, the `e1071` package is commonly employed.

Gradient Boosting Classifier

Boosting algorithms work by sequentially combining multiple weak learners—models that perform slightly better than random guessing—into a more powerful, collective learner. This process occurs in an iterative manner, progressively improving the model's performance. Gradient boosting, a specific type of boosting algorithm, is used for regression tasks and applies this iterative strategy to build a robust predictive model by minimizing errors step by step [4]. This method is notably effective for regression tasks, enhancing model performance by reducing errors step by step. Despite its strong predictive power, gradient boosting can be computationally demanding and susceptible to overfitting without appropriate regularization. The prediction formula for gradient boosting is:

$$F(x) = \sum_{m=1}^{M} \gamma_m h_m(x)$$

where $F(x)$ represents the final prediction, $h_m(x)$ are the weak learners, and γ_m is the learning rate [4]. Popular implementation packages include Python's `scikit-learn` and `xgboost`, as well as R's `gbm` and `xgboost`.

Naive Bayes

Naive Bayes is a probabilistic classification technique based on Bayes' theorem, which assumes that features are conditionally independent given the class label. This assumption of independence simplifies the computation of probabilities, making Naive Bayes computationally efficient and easy to implement. Despite its simplicity, Naive Bayes performs remarkably well in domains like text classification and spam detection, where its assumptions are often approximately valid [33].
The simplicity of Naive Bayes comes with limitations. The assumption of feature independence may not hold in practice, which can lead to suboptimal performance in datasets where features are strongly correlated. Additionally, Naive Bayes is less effective with continuous features unless they are discretized or transformed using techniques like Gaussian or kernel density estimation.
The posterior probability in Naive Bayes is calculated using Bayes' theorem:

$$P(y \mid \mathbf{x}) = \frac{P(y) \prod_{i=1}^{n} P(x_i \mid y)}{P(\mathbf{x})}$$

where $P(y \mid \mathbf{x})$ is the posterior probability of class y given the input feature vector $\mathbf{x}$, $P(y)$ is the prior probability of the class, $P(x_i \mid y)$ is the likelihood of feature x_i given the class, and $P(\mathbf{x})$ is the evidence, which acts as a normalizing constant.
The classifier assigns the class y that maximizes the posterior probability:

$$\hat{y} = \arg\max_{y} P(y) \prod_{i=1}^{n} P(x_i \mid y)$$

Naive Bayes is widely supported in various programming packages. In Python, it can be implemented using the `scikit-learn` library, which offers Gaussian, Multinomial, and Bernoulli Naive Bayes variants. In R, common implementations include the `e1071` and `naivebayes` packages, which provide versatile tools for building Naive Bayes classifiers.

Random Forest

Random Forest, introduced by Breiman and Cutler, is an ensemble learning method that is widely used for both classification and regression tasks. It constructs multiple decision trees during training and aggregates their predictions to improve accuracy and robustness. The final output is determined by majority voting in classification tasks or by averaging the predictions in regression tasks. This method is particularly valued for its high predictive accuracy, ability to handle large datasets with missing values, and integrated feature importance evaluation [24].
One of the primary advantages of Random Forest is its resistance to overfitting, especially when compared to single decision trees. Additionally, it performs well on datasets with high dimensionality and complex feature interactions. However, its ensemble nature can make it less interpretable than individual decision trees, and it may become computationally intensive when a large number of trees are used.
The mathematical formulation of Random Forest combines predictions from T decision trees $h_t(x)$, as follows:

$$F(x) = \frac{1}{T} \sum_{t=1}^{T} h_t(x)$$

where $h_t(x)$ represents the prediction from the t-th decision tree, and T is the total number of trees in the forest. The method introduces randomness in two ways: by bootstrapping samples for each tree and by randomly selecting subsets of features at each split, which helps reduce overfitting and enhances generalization.
Random Forest can be implemented using various programming tools. In Python, the `scikit-learn` library provides an efficient implementation, while in R, the `randomForest` package is commonly employed [24].

K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a simple and intuitive classification technique that assigns a label

to an unlabeled data point based on the majority class among its k-nearest neighbors in the feature space. This approach assumes that data points close to one another are likely to belong to the same class, making it particularly effective for tasks where local proximity-based classification is relevant [35]. The choice of k, the number of neighbors considered, significantly influences the model's performance, balancing between overfitting (k too small) and underfitting (k too large).
KNN is non-parametric and makes no assumptions about the underlying data distribution, which enhances its versatility across a variety of datasets. However, it can be computationally intensive during prediction, as it requires calculating distances to all training points for each test instance. Additionally, KNN is sensitive to irrelevant features and the scaling of data, requiring careful preprocessing.
The classification decision in KNN is determined by:

$$\hat{y} = \text{mode}(\{y_i : x_i \in \text{k-NN}(\mathbf{x})\})$$

where $\hat{y}$ is the predicted label for the input $\mathbf{x}$, y_i represents the labels of the k-nearest neighbors x_i, and k-NN($\mathbf{x}$) denotes the set of k-nearest neighbors to $\mathbf{x}$ in the feature space. Distance metrics such as Euclidean, Manhattan, or Minkowski are typically used to compute proximity.
KNN can be implemented using a variety of programming tools. In Python, the `scikit-learn` library provides a straightforward implementation. In R, the `class` package is commonly employed for KNN-based modeling [35].

XGBoost

Extreme Gradient Boosting (XGBoost) is a high-performance, scalable framework specifically designed for gradient tree boosting. It has gained widespread recognition for its efficiency and superior predictive capabilities, especially in competitive machine learning tasks. By incorporating advanced optimization techniques, XGBoost enhances the traditional gradient boosting algorithm, making it a go-to tool for predictive modeling across various domains [8].
XGBoost combines boosting principles with regularization to prevent overfitting and improve model generalization. It builds a sequence of weak learners (decision trees), where each tree corrects the errors of its predecessors. Despite its effectiveness, XGBoost requires careful tuning of hyperparameters, such as the learning rate and the number of trees, to achieve optimal performance. Additionally, its computational complexity can be a concern for very large datasets.
The objective of XGBoost is to minimize a regularized loss function, defined as:

$$L = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

where $\ell(y_i, \hat{y}_i)$ represents the loss function (e.g., mean squared error for regression or log loss for classification), and $\Omega(f_k)$ is the regularization term for the complexity of the model. The regularization term Ω helps control overfitting by penalizing more complex models. Each tree f_k is added iteratively to the model to minimize this loss.
XGBoost provides flexible implementation options, supporting sparse data, customized objective functions, and parallel computing. It can be implemented using widely available programming libraries, including:

- Python: `xgboost`
- R: `xgboost`

Chapter 5

Data Pre-processing and Experimental Settings

5.1 Preprocessing

Preprocessing is a critical step in preparing data for machine learning models, ensuring the dataset is clean, consistent, and ready for analysis. This section details the comprehensive preprocessing procedures applied to the datasets used in this study, aimed at maximizing the integrity and utility of the data for predictive modeling.

The initial step involved addressing **missing values**, which can introduce bias or reduce the dataset's representativeness. Numerical features with missing entries were imputed using their mean values, ensuring minimal disruption to data distribution. Categorical variables were filled with their mode, maintaining the representativeness of dominant categories. In cases of extensive missingness, rows or columns were excluded after assessing their significance to avoid diluting the dataset's quality.

Outliers were managed to prevent their disproportionate impact on model performance. Statistical methods such as the interquartile range (IQR) were employed to identify extreme values. Depending on their severity, outliers were either capped to boundary values or transformed using logarithmic scaling, reducing their influence while preserving overall patterns.

Data Normalization was applied to numerical variables to ensure features with differing scales contributed equitably to model training. The min-max scaling technique was adopted, transforming all numerical values into a uniform range of [0,1]. This step was particularly important for algorithms sensitive to magnitude differences, such as gradient-based optimizers.

A notable challenge in the dataset was **class imbalance**, particularly across the four target classification tasks: complications binary, complications multi-class, relapses binary, and relapses multi-class. The Synthetic Minority Oversampling Technique (SMOTE) was employed to balance the class distributions by generating synthetic samples for underrepresented classes. This approach improved the machine learning models' ability to generalize across all classes.

The dataset was then divided into **training and testing splitting** to evaluate model performance effectively. A stratified sampling strategy was implemented to preserve the original class distributions within both sets, ensuring consistency in evaluation metrics.

Finally, comprehensive data validation was performed to confirm the consistency of variable types,

ranges, and formats. Any discrepancies were resolved before proceeding to feature engineering and model building. This robust preprocessing pipeline laid a strong foundation for the subsequent machine learning analysis, enhancing model reliability and predictive performance.

5.2 Experimental Settings

This study involves four classification tasks, each designed to address distinct aspects of post-operative outcomes following hemorrhoid surgery. These tasks focus on two target variables: complications and relapses. For each variable, we consider both binary and multiclass classification, resulting in the following four tasks:

1. **Complication Binary Classification:** Predicting the presence (yes/no) of post-operative complications.
2. **Complication Multiclass Classification:** Predicting the severity of complications, categorized into multiple classes.
3. **Relapse Binary Classification:** Predicting the occurrence (yes/no) of post-operative relapses.
4. **Relapse Multiclass Classification:** Predicting the type or stage of relapse, categorized into multiple classes.

Train-Test Splitting: To evaluate the performance of the machine learning models, the dataset was split into training and testing subsets. A standard train-test split of 80%-20% was applied, ensuring that the distribution of classes in the target variable remained consistent between the subsets. This split helps prevent data leakage and provides an unbiased assessment of the models' generalization performance.

Validation Strategy: Grid search with cross-validation was employed to tune hyperparameters for each model. A 5-fold cross-validation strategy was used during grid search to ensure that the models were validated on different subsets of the training data. This approach helps mitigate overfitting and ensures robust hyperparameter selection. The grid search process involved systematically testing combinations of hyperparameters and selecting the set that yielded the best performance on the validation folds. Metrics such as balanced accuracy, precision, recall, and F1-score were used to evaluate model performance during the validation phase.

Preprocessing Considerations: Prior to splitting and model training, the dataset underwent extensive preprocessing. Missing values were imputed using suitable methods, and class imbalance in the binary and multiclass targets was addressed using the Synthetic Minority Oversampling Technique (SMOTE). Data normalization was applied to ensure consistency and compatibility across features for algorithms sensitive to scale.

The experimental setup ensures a fair and reproducible comparison of different machine learning algorithms for each of the four tasks. This systematic approach allows us to identify the most effective models and configurations for predicting post-operative outcomes.

5.3 Data Imbalance Handling

Data imbalance is a significant challenge in classification tasks, where one class may have significantly fewer samples than others. This imbalance can lead to biased models that perform poorly

on the minority class. To address this issue, the Synthetic Minority Over-sampling Technique (SMOTE) was employed, and careful training and test splitting strategies were applied.

SMOTE is a widely used method to handle class imbalance by generating synthetic samples for the minority class. It creates new samples by interpolating between existing ones in the feature space, effectively increasing the representation of the minority class without introducing duplicates. This process ensures that the dataset is more balanced, improving the model's ability to generalize across all classes.

The implementation of SMOTE involves the following steps:

- For each sample in the minority class, identify its k-nearest neighbors.
- Randomly select one of these neighbors.
- Generate a synthetic sample by linear interpolation between the selected sample and its neighbor:

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda(\mathbf{x}_j - \mathbf{x}_i)$$

  where $\mathbf{x}_i$ and $\mathbf{x}_j$ are the feature vectors of the original sample and its neighbor, respectively, and λ is a random number in the range $[0, 1]$.

In addition to SMOTE, the dataset was carefully split into training and test sets to ensure fair evaluation of the model's performance. The splitting process followed these principles:

- The minority class was proportionally represented in both training and test sets to prevent bias in model evaluation.
- Stratified sampling was applied during splitting to maintain the same class distribution across subsets.
- A typical split ratio of 80% for training and 20% for testing was adopted, ensuring sufficient data for training while reserving a significant portion for evaluation.

These techniques collectively ensured that the model was trained on a balanced dataset and evaluated fairly, leading to improved performance and robustness in predicting minority class instances.

Chapter 6

Results

6.1 Preliminary Results

This section presents an analysis of various machine learning models applied to the target variables: **Complication Binary**, **Relapses Binary**, **Complication Multi-class**, and **Relapses Multi-class**. The evaluation is conducted on the original imbalanced datasets and after applying different levels of class re-balancing. The models assessed include Logistic Regression, Gradient Boosting Classifier, KNN, SVM, Naive Bayes, Random Forest, and XGBoost. Performance metrics such as Balanced Accuracy, Precision, Recall, F1-score, and Support are used to evaluate each model's effectiveness.

Class re-balancing techniques were applied incrementally, with re-balanced class distributions ranging from 40 percent to 100 percent, to examine the impact of balancing on model performance. However, our experiments revealed significant challenges with the multi-class target variables **(Complication Multi-class and Relapses Multi-class)**. Due to severe class imbalances, where certain classes were underrepresented, the models struggled to provide accurate predictions. This imbalance led to poor performance across most metrics, particularly for underrepresented classes. Attempts to reduce the number of classes and simplify the problem failed to yield meaningful improvements, suggesting that the dataset's structure inherently limits effective prediction for the multi-class targets.

Consequently, we shifted our focus to the binary target variables: **Complication Binary** and **Relapses Binary**. These tasks yield more stable and interpretable models, highlighting the importance of addressing class imbalances and dataset characteristics in predictive modeling. This pivot allowed us to concentrate on extracting actionable insights from the binary classification tasks, where the models showed more promise.

6.2 Complication Binary

In this section, we describe the steps taken to analyze and model the **Complication Binary** target variable. The process began with the application of essential pre-processing techniques to clean and prepare the dataset for analysis. The data was then split into training and testing sets to ensure unbiased evaluation of model performance. Given the presence of class imbalance in the dataset, we employed the SMOTE (Synthetic Minority Over-sampling Technique) algorithm to rebalance the

classes at various ratio levels.
The impact of class rebalancing was systematically assessed by comparing model performance on the original imbalanced dataset with that on the rebalanced datasets. For each model, we tracked key performance metrics, including Balanced Accuracy, Precision, Recall, F1-score, and Support. This comprehensive evaluation provided insights into how rebalancing influenced the ability of the models to generalize across both the minority and majority classes.
To further optimize performance, we utilized a Grid Search approach to identify the best hyperparameters for each model. By systematically exploring combinations of parameters, we fine-tuned the models to achieve their maximum potential. The results were analyzed to determine whether class rebalancing and hyperparameter optimization improved the predictive capabilities of the models. This step-by-step approach ensured a robust evaluation of the **Complication Binary** target variable and highlighted the importance of addressing class imbalance and parameter tuning in achieving accurate and reliable predictions.

Key Observations:

- **Logistic Regression**:
  - **Imbalanced Dataset**: Shows high precision (85, 83) but extremely low recall (100, 9) for the minority class, indicating a strong bias towards the majority class.
  - **Rebalanced Dataset**: As the rebalancing ratio increases from 65% to 100%, the balanced accuracy fluctuates but generally improves compared to the imbalanced dataset. Precision and recall for the minority class also improve , peaking around the 75% rebalancing ratio, suggesting that a moderate level of oversampling provides the best balance between precision and recall.
- **Gradient Boosting Classifier**:
  - **Imbalanced Dataset**: Consistently high balanced accuracy (74) and strong precision (91, 93), but lower recall for the minority class (99, 49).
  - **Rebalanced Dataset**: Balanced accuracy remains fairly stable (around 71-74). Precision and recall for the minority class show slight improvements, indicating that while rebalancing helps, the model's performance is already robust.
- **KNN**:
  - **Imbalanced Dataset**: Decent balanced accuracy (71) and precision (90, 81), but low recall (98, 45) for the minority class.
  - **Rebalanced Dataset**: Balanced accuracy fluctuates with rebalancing (67-70). Precision and recall for the minority class show some improvement but remain inconsistent, highlighting that KNN struggles more with changes in class distribution.
- **SVM**:
  - **Imbalanced Dataset**: Good balanced accuracy (73) and precision (91, 82), with low recall (98, 49) for the minority class.
  - **Rebalanced Dataset**: Balanced accuracy and recall for the minority class improve with rebalancing, peaking around 75%. This suggests that SVM benefits significantly from moderate rebalancing, achieving a better trade-off between precision and recall.
- **Naive Bayes**:
  - **Imbalanced Dataset**: Moderate balanced accuracy (68) with high precision (92, 30)

but poor recall (70, 67).
  - **Rebalanced Dataset**: Balanced accuracy decreases as rebalancing ratios increase, indicating challenges in handling rebalanced data. Precision and recall for the minority class fluctuate, showing Naive Bayes is less effective with higher levels of oversampling.
- **Random Forest**:
  - **Imbalanced Dataset**: High balanced accuracy (74) and precision (91, 93), but low recall (99, 49) for the minority class.
  - **Rebalanced Dataset**: Balanced accuracy remains relatively stable (around 70-74), with minor improvements in recall and balanced accuracy. This shows that Random Forest is robust across different rebalancing ratios.
- **XGBoost**:
  - **Imbalanced Dataset**: Good balanced accuracy (71) and precision (91, 83), but low recall (98, 45) for the minority class.
  - **Rebalanced Dataset**: Balanced accuracy remains stable (around 70-72). Precision and recall for the minority class improve with higher rebalancing ratios, maintaining consistent performance and indicating the model's adaptability to rebalanced data.

Further Observations:

In analyzing the models, we observed several key trends and trade-offs. Most models show high precision for the majority class but very low recall for the minority class in the imbalanced dataset, indicating difficulty in identifying complications. When the dataset is rebalanced, increasing rebalancing ratios generally improve recall for the minority class, though sometimes at the expense of a slight decrease in precision. This suggests a trade-off between identifying more true positives (recall) and minimizing false positives (precision).
Balanced accuracy is generally lower for most models in the imbalanced dataset due to significant class imbalance. However, balanced accuracy improves across most models with increased rebalancing ratios, indicating that the models become better at handling the minority class without losing too much accuracy on the majority class.
Gradient Boosting Classifier and Random Forest models exhibit consistent performance and are robust across different rebalancing ratios. Support Vector Machine shows significant improvement with moderate rebalancing, achieving a good balance between precision and recall. On the other hand, Naive Bayes struggles with high rebalancing ratios, leading to decreased performance.
Performance for Logistic Regression and SVM models tends to peak around the 75% rebalancing ratio, suggesting an optimal level of rebalancing where these models achieve the best trade-off between precision and recall. KNN shows more variability, indicating sensitivity to changes in class distribution.
XGBoost demonstrates consistent performance and slight improvements in recall and balanced accuracy, making it suitable for predictive modeling in imbalanced datasets. This adaptability, combined with its scalability and speed, makes it a strong candidate for handling class imbalance effectively.

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 54 Precision: 85 ; 83 Recall: 100 ; 9 f1-score: 92 ; 16 Support: 291 ; 55	Balance Accuracy: 73 Precision: 91 ; 77 Recall: 97 ; 49 f1-score: 94 ; 60 Support: 291 ; 55
Gradient Boosting Classifier	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 73 Precision: 91 ; 84 Recall: 98 ; 49 f1-score: 95 ; 62 Support: 291 ; 55
KNN	Balance Accuracy: 71 Precision: 90 ; 81 Recall: 98 ;45 f1-score: 94 ; 58 Support: 291 ; 55	Balance Accuracy: 67 Precision: 90 ; 41 Recall: 87 ; 49 f1-score: 88 ; 45 Support: 291 ; 55
Support vector machine	Balance Accuracy: 73 Precision: 91 ; 82 Recall: 98 ; 49 f1-score: 94 ; 61 Support: 291 ; 55	Balance Accuracy: 75 Precision: 92 ; 64 Recall: 94 ; 58 f1-score: 93 ; 61 Support: 291 ; 55
Naive Bayes	Balance Accuracy: 68 Precision: 92 ; 30 Recall: 70 ; 67 f1-score: 80 ; 41 Support: 291 ; 55	Balance Accuracy: 61 Precision: 91 ; 21 Recall:46 ; 76 f1-score: 62 ; 33 Support: 291 ;55
Random Forest	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 71 Precision: 90 ; 78 Recall: 98 ; 45 f1-score: 94 ; 57 Support: 291 ; 55
XGboost	Balance Accuracy: 71 Precision: 91 ; 83 Recall: 98 ; 45 f1-score: 94 ; 59 Support: 291 ; 55	Balance Accuracy: 71 Precision: 91 ; 47 Recall: 88 ; 55 f1-score:90 ; 51 Support: 291 ; 55

Table 6.1: Complication Binary with 40 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 54 Precision: 85 ; 83 Recall: 100 ; 9 f1-score: 92 ; 16 Support: 291 ; 55	Balance Accuracy: 73 Precision: 91 ; 76 Recall: 97 ; 51 f1-score: 94 ; 61 Support: 291 ; 55
Gradient Boosting Classifier	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 72 Precision: 91 ; 66 Recall: 95 ; 49 f1-score: 93 ; 56 Support: 291 ; 55
KNN	Balance Accuracy: 71 Precision: 90 ; 81 Recall: 98 ;45 f1-score: 94 ; 58 Support: 291 ; 55	Balance Accuracy: 67 Precision: 90 ; 40 Recall: 86 ; 49 f1-score: 88 ; 44 Support: 291 ; 55
Support vector machine	Balance Accuracy: 73 Precision: 91 ; 82 Recall: 98 ; 49 f1-score: 94 ; 61 Support: 291 ; 55	Balance Accuracy: 75 Precision: 92 ; 64 Recall: 94 ; 58 f1-score: 93 ; 61 Support: 291 ; 55
Naive Bayes	Balance Accuracy: 68 Precision: 92 ; 30 Recall: 70 ; 67 f1-score: 80 ; 41 Support: 291 ; 55	Balance Accuracy: 60 Precision: 91 ; 20 Recall: 40 ; 80 f1-score: 56 ; 32 Support: 291 ; 55
Random Forest	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 70 Precision: 90 ; 62 Recall: 95 ; 45 f1-score: 92 ; 53 Support: 291 ; 55
XGboost	Balance Accuracy: 71 Precision: 91 ; 83 Recall: 98 ; 45 f1-score: 94 ; 59 Support: 291 ; 55	Balance Accuracy: 70 Precision: 91 ; 47 Recall: 89 ; 51 f1-score: 90 ; 49 Support: 291 ; 55

Table 6.2: Complication Binary with 50 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 54 Precision: 85 ; 83 Recall: 100 ; 9 f1-score: 92 ; 16 Support: 291 ; 55	Balance Accuracy: 74 Precision: 92 ;72 Recall: 96 ; 53 f1-score: 94 ; 61 Support: 291 ; 55
Gradient Boosting Classifier	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55
KNN	Balance Accuracy: 71 Precision: 90 ; 81 Recall: 98 ;45 f1-score: 94 ; 58 Support: 291 ; 55	Balance Accuracy: 70 Precision: 91 ; 45 Recall: 88 ; 53 f1-score: 89 ; 48 Support: 291 ; 55
Support vector machine	Balance Accuracy: 73 Precision: 91 ; 82 Recall: 98 ; 49 f1-score: 94 ; 61 Support: 291 ; 55	Balance Accuracy: 75 Precision: 92 ; 64 Recall: 94 ; 58 f1-score: 93 ; 61 Support: 291 ; 55
Naive Bayes	Balance Accuracy: 68 Precision: 92 ; 30 Recall: 70 ; 67 f1-score: 80 ; 41 Support: 291 ; 55	Balance Accuracy: 59 Precision: 91 ; 20 Recall: 39 ; 80 f1-score: 54 ; 32 Support: 291 ; 55
Random Forest	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 69 Precision: 90 ; 60 Recall: 94 ; 45 f1-score: 92 ; 52 Support: 291 ; 55
XGboost	Balance Accuracy: 71 Precision: 91 ; 83 Recall: 98 ; 45 f1-score: 94 ; 59 Support: 291 ; 55	Balance Accuracy: 71 Precision: 91 ; 53 Recall: 91 ; 53 f1-score: 91 ;53 Support: 291 ; 55

Table 6.3: Complication Binary with 60 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 54 Precision: 85 ; 83 Recall: 100 ; 9 f1-score: 92 ; 16 Support: 291 ; 55	Balance Accuracy:74 Precision: 92 ; 64 Recall: 94 ; 55 f1-score: 93 ; 59 Support: 291 ; 55
Gradient Boosting Classifier	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 72 Precision: 91 ; 65 Recall: 95 ; 51 f1-score: 93 ; 57 Support: 291 ; 55
KNN	Balance Accuracy: 71 Precision: 90 ; 81 Recall: 98 ;45 f1-score: 94 ; 58 Support: 291 ; 55	Balance Accuracy: 68 Precision: 90 ; 39 Recall: 85 ; 53 f1-score: 87 ; 45 Support: 291 ; 55
Support vector machine	Balance Accuracy: 73 Precision: 91 ; 82 Recall: 98 ; 49 f1-score: 94 ; 61 Support: 291 ; 55	Balance Accuracy: 75 Precision: 92 ; 64 Recall: 94 ; 58 f1-score: 93 ; 61 Support: 291 ; 55
Naive Bayes	Balance Accuracy: 68 Precision: 92 ; 30 Recall: 70 ; 67 f1-score: 80 ; 41 Support: 291 ; 55	Balance Accuracy: 58 Precision: 91 ; 19 Recall: 37 ; 80 f1-score: 52 ; 31 Support: 291 ; 55
Random Forest	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 70 Precision: 91 ; 47 Recall: 89 ; 51 f1-score: 90 ; 49 Support: 291 ; 55
XGboost	Balance Accuracy: 71 Precision: 91 ; 83 Recall: 98 ; 45 f1-score: 94 ; 59 Support: 291 ; 55	Balance Accuracy: 70 Precision: 91 ; 47 Recall: 89 ; 51 f1-score: 90 ; 49 Support: 291 ; 55

Table 6.4: Complication Binary with 65 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 54 Precision: 85 ; 83 Recall: 100 ; 9 f1-score: 92 ; 16 Support: 291 ; 55	Balance Accuracy: 75 Precision:93 ;52 Recall: 89 ;62 f1-score: 91 ; 56 Support: 291 ; 55
Gradient Boosting Classifier	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 72 Precision: 91 ; 66 Recall: 95 ; 49 f1-score: 93 ; 56 Support: 291 ; 55
KNN	Balance Accuracy: 71 Precision: 90 ; 81 Recall: 98 ;45 f1-score: 94 ; 58 Support: 291 ; 55	Balance Accuracy: 70 Precision: 91 ; 43 Recall: 86 ; 55 f1-score: 89 ; 48 Support: 291 ; 55
Support vector machine	Balance Accuracy: 73 Precision: 91 ; 82 Recall: 98 ; 49 f1-score: 94 ; 61 Support: 291 ; 55	Balance Accuracy: 75 Precision: 92 ; 63 Recall: 94 ; 56 f1-score: 93 ; 60 Support: 291 ; 55
Naive Bayes	Balance Accuracy: 68 Precision: 92 ; 30 Recall: 70 ; 67 f1-score: 80 ; 41 Support: 291 ; 55	Balance Accuracy: 57 Precision: 90 ; 19 Recall: 35 ; 80 f1-score: 51 ; 31 Support: 291 ; 55
Random Forest	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 68 Precision: 90 ; 53 Recall: 92 ; 45 f1-score: 91 ; 49 Support: 291 ; 55
XGboost	Balance Accuracy: 71 Precision: 91 ; 83 Recall: 98 ; 45 f1-score: 94 ; 59 Support: 291 ; 55	Balance Accuracy: 71 Precision: 91 ; 50 Recall: 90 ; 53 f1-score: 91 ; 51 Support: 291 ; 55

Table 6.5: Complication Binary with 75 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 54 Precision: 85 ; 83 Recall: 100 ; 9 f1-score: 92 ; 16 Support: 291 ; 55	Balance Accuracy: 71 Precision: 93 ; 35 Recall: 76 ; 67 f1-score: 84 ; 46 Support: 291 ; 55
Gradient Boosting Classifier	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 71 Precision: 91 ; 59 Recall: 93 ; 49 f1-score: 92 ; 53 Support: 291 ; 55
KNN	Balance Accuracy: 71 Precision: 90 ; 81 Recall: 98 ;45 f1-score: 94 ; 58 Support: 291 ; 55	Balance Accuracy: 67 Precision: 90 ; 35 Recall: 81 ; 53 f1-score: 86 ; 42 Support: 291 ; 55
Support vector machine	Balance Accuracy: 73 Precision: 91 ; 82 Recall: 98 ; 49 f1-score: 94 ; 61 Support: 291 ; 55	Balance Accuracy: 73 Precision: 92 ; 46 Recall: 87 ; 60 f1-score: 89 ; 52 Support: 291 ; 55
Naive Bayes	Balance Accuracy: 68 Precision: 92 ; 30 Recall: 70 ; 67 f1-score: 80 ; 41 Support: 291 ; 55	Balance Accuracy: 57 Precision: 90 ; 19 Recall: 34 ; 80 f1-score: 50 ; 30 Support: 291 ; 55
Random Forest	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 69 Precision: 90 ; 48 Recall: 90 ; 49 f1-score: 90 ; 49 Support: 291 ; 55
XGboost	Balance Accuracy: 71 Precision: 91 ; 83 Recall: 98 ; 45 f1-score: 94 ; 59 Support: 291 ; 55	Balance Accuracy: 71 Precision: 91 ; 63 Recall: 95 ; 49 f1-score: 93 ; 55 Support: 291 ; 55

Table 6.6: Complication Binary with 85 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 54 Precision: 85 ; 83 Recall: 100 ; 9 f1-score: 92 ; 16 Support: 291 ; 55	Balance Accuracy: 65 Precision: 91 ; 25 Recall: 59 ; 71 f1-score: 72 ; 37 Support: 291 ; 55
Gradient Boosting Classifier	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 71 Precision: 91 ; 61 Recall: 94 ; 49 f1-score: 92 ; 55 Support: 291 ; 55
KNN	Balance Accuracy: 71 Precision: 90 ; 81 Recall: 98 ;45 f1-score: 94 ; 58 Support: 291 ; 55	Balance Accuracy: 69 Precision: 91 ;39 Recall: 84 ; 55 f1-score: 87 ; 45 Support: 291 ; 55
Support vector machine	Balance Accuracy: 73 Precision: 91 ; 82 Recall: 98 ; 49 f1-score: 94 ; 61 Support: 291 ; 55	Balance Accuracy: 67 Precision: 91 ; 29 Recall: 70 ; 65 f1-score: 79 ; 40 Support: 291 ; 55
Naive Bayes	Balance Accuracy: 68 Precision: 92 ; 30 Recall: 70 ; 67 f1-score: 80 ; 41 Support: 291 ; 55	Balance Accuracy: 55 Precision: 90 ; 18 Recall: 30 ; 82 f1-score: 45 ; 30 Support: 291 ; 55
Random Forest	Balance Accuracy: 74 Precision: 91 ; 93 Recall: 99 ; 49 f1-score: 95 ; 64 Support: 291 ; 55	Balance Accuracy: 70 Precision: 91 ; 45 Recall: 88 ; 53 f1-score: 89 ; 48 Support: 291 ; 55
XGboost	Balance Accuracy: 71 Precision: 91 ; 83 Recall: 98 ; 45 f1-score: 94 ; 59 Support: 291 ; 55	Balance Accuracy: 72 Precision: 91 ; 65 Recall: 95 ; 51 f1-score: 93 ; 57 Support: 291 ; 55

Table 6.7: Complication Binary with 100 percent re-balance class

6.3 Relapses Binary

This section outlines the modeling and analysis process for the second target variable, **Relapses Binary**.To ensure the dataset was well-prepared for analysis, we began with comprehensive preprocessing steps. The data was subsequently split into training and testing sets, ensuring an unbiased evaluation framework for all models.
As with the **Complication Binary** target, the **Relapses Binary** dataset exhibited significant class imbalance. To address this issue, we employed the SMOTE (Synthetic Minority Over-sampling Technique) algorithm to generate synthetic samples for the minority class. This rebalancing was performed at various ratio levels, enabling us to systematically examine its impact on model performance.
We evaluated the models using key performance metrics such as Balanced Accuracy, Precision, Recall, F1-score, and Support, comparing their performance on the original imbalanced dataset with that on the rebalanced datasets. This comparison provided valuable insights into the extent to which rebalancing improved the models' ability to handle the underrepresented class.
Additionally, a Grid Search approach was employed to fine-tune the hyperparameters of each model. By optimizing parameters, we sought to enhance the predictive accuracy and robustness of the models. The combination of class rebalancing and parameter optimization ensured a comprehensive evaluation of the "Relapses Binary" target variable. The findings underscore the importance of addressing class imbalance and selecting optimal configurations to improve predictive performance in imbalanced datasets.

Key Observations:

- **Logistic Regression**:
  - **Imbalanced Dataset**: Shows high precision (90, 0) for the majority class but extremely low recall (100, 0) for the minority class, indicating a strong bias towards the majority class.
  - **Rebalanced Dataset**: As the rebalancing ratio increases from 65% to 100%, balanced accuracy and recall for the minority class improve significantly. Precision for the minority class also improves, with the performance peaking around 75% rebalancing. Higher rebalancing ratios maintain improved recall, but precision can fluctuate.
- **Gradient Boosting Classifier**:
  - **Imbalanced Dataset**: Consistently moderate balanced accuracy (51) and good precision for the majority class (91, 25) but low recall (99, 4) for the minority class.
  - **Rebalanced Dataset**: Balanced accuracy and recall for the minority class improve with increasing rebalancing ratios, peaking around 85% to 100%. Precision for the minority class improves, indicating effective handling of class imbalance.
- **KNN**:
  - **Imbalanced Dataset**: Shows high precision for the majority class (91, 0) but zero recall (100, 0) for the minority class, indicating poor performance on minority class predictions.
  - **Rebalanced Dataset**: Balanced accuracy and recall for the minority class improve significantly with increasing rebalancing ratios, peaking around 75% to 85%. Precision for the minority class improves but shows some variability, suggesting sensitivity to changes

in class distribution.

- **SVM**:
  - **Imbalanced Dataset**: Good balanced accuracy (50) with high precision for the majority class (91, 0) but zero recall (100, 0) for the minority class.
  - **Rebalanced Dataset**: Balanced accuracy and recall for the minority class improve with increasing rebalancing ratios, peaking around 75% to 85%. This indicates that SVM benefits from rebalancing, achieving a better trade-off between precision and recall.
- **Naive Bayes**:
  - **Imbalanced Dataset**: Shows moderate balanced accuracy (57) with high precision for the majority class (98, 11) but very low recall (19, 96) for the minority class and high recall for the majority class.
  - **Rebalanced Dataset**: Performance remains relatively stable with no significant improvements across different rebalancing ratios, indicating challenges in handling rebalanced data.
- **Random Forest**:
  - **Imbalanced Dataset**: Shows moderate balanced accuracy (52) with high precision (91) but very low recall (100, 4) for the minority class.
  - **Rebalanced Dataset**: Balanced accuracy and recall for the minority class improve with increasing rebalancing ratios, peaking around 85% to 100%. Precision for the minority class also improves, indicating better handling of class imbalance.
- **XGBoost**:
  - **Imbalanced Dataset**: Shows high precision for the majority class (91, 14) but very low recall (97, 4) for the minority class.
  - **Rebalanced Dataset**: Balanced accuracy and recall for the minority class improve with increasing rebalancing ratios, peaking around 85% to 100%. Precision for the minority class also shows some improvement, indicating better adaptability to rebalanced data.

Further Observations:

In analyzing the models, we observed several key trends and trade-offs. Most models show high precision for the majority class but very low recall for the minority class in the imbalanced dataset, indicating difficulty in identifying relapses. When the dataset is rebalanced, increasing rebalancing ratios generally improve recall for the minority class. However, this often comes with a slight decrease in precision, suggesting a trade-off between identifying more true positives (recall) and minimizing false positives (precision).
Balanced accuracy is generally lower for most models in the imbalanced dataset due to significant class imbalance. However, balanced accuracy improves across most models as the rebalancing ratio increases, indicating that the models become better at handling the minority class without losing too much accuracy on the majority class.
Gradient Boosting Classifier and Random Forest models exhibit consistent performance across different rebalancing ratios, maintaining high balanced accuracy and reasonable trade-offs between precision and recall. This robustness makes them suitable for handling imbalanced datasets. Support Vector Machine shows significant improvement with higher rebalancing ratios, achieving a good

balance between precision and recall, making it a viable option for imbalanced medical datasets. On the other hand, Naive Bayes struggles with higher rebalancing ratios, leading to decreased performance. This model may not be the best choice for highly imbalanced datasets unless the class distribution is carefully managed.

Performance for Logistic Regression and SVM models tends to peak around the 75% to 85% rebalancing ratios, suggesting an optimal level of rebalancing where these models achieve the best trade-off between precision and recall. KNN shows more variability, indicating sensitivity to changes in class distribution. This model may require additional tuning or alternative techniques to handle class imbalance effectively.

XGBoost demonstrates consistent performance across different rebalancing ratios, with slight improvements in recall and balanced accuracy. This adaptability, combined with its scalability and speed, makes it a strong candidate for predictive modeling in imbalanced datasets.

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 50 Precision: 90 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 53 Precision: 91 ; 19 Recall: 94 ; 13 f1-score: 93 ; 15 Support: 227 ; 23
Gradient Boosting Classifier	Balance Accuracy: 51 Precision: 91 ; 25 Recall: 99 ; 4 f1-score: 95 ; 7 Support: 227 ; 23	Balance Accuracy: 56 Precision: 92 ; 21 Recall: 92 ; 22 f1-score: 92 ;21 Support: 227 ; 23
KNN	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 58 Precision: 92 ; 18 Recall: 86 ; 30 f1-score: 89 ; 23 Support: 227 ; 23
Support vector machine	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 56 Precision: 92 ; 25 Recall: 95 ; 17 f1-score: 93 ; 21 Support: 227 ; 23
Naive Bayes	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23
Random Forest	Balance Accuracy: 52 Precision: 91 ; 100 Recall: 100 ; 4 f1-score: 95 ; 8 Support: 227 ; 23	Balance Accuracy: 54 Precision: 92 ; 19 Recall: 93 ; 17 f1-score: 92 ; 18 Support: 227 ; 23
XGboost	Balance Accuracy: 50 Precision: 91 ; 14 Recall: 97 ; 4 f1-score: 94 ; 7 Support: 227 ; 23	Balance Accuracy: 48 Precision: 91 ; 0 Recall: 97 ; 0 f1-score: 94 ; 0 Support: 227 ; 23

Table 6.8: Relapses Binary with 40 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 50 Precision: 90 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 53 Precision: 91 ; 14 Recall: 89 ; 17 f1-score: 90 ; 15 Support: 227 ; 23
Gradient Boosting Classifier	Balance Accuracy: 51 Precision: 91 ; 25 Recall: 99 ; 4 f1-score: 95 ; 7 Support: 227 ; 23	Balance Accuracy: 47 Precision: 90 ; 0 Recall: 95 ; 0 f1-score: 92 ; 0 Support: 227 ; 23
KNN	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 62 Precision: 93 ; 21 Recall: 85 ; 39 f1-score: 89 ; 27 Support: 227 ; 23
Support vector machine	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 54 Precision: 92 ; 17 Recall: 91 ; 17 f1-score: 91 ; 17 Support: 227 ; 23
Naive Bayes	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23
Random Forest	Balance Accuracy: 52 Precision: 91 ; 100 Recall: 100 ; 4 f1-score: 95 ; 8 Support: 227 ; 23	Balance Accuracy: 56 Precision: 92 ; 19 Recall: 91 ; 22 f1-score: 91 ; 20 Support: 227 ; 23
XGboost	Balance Accuracy: 50 Precision: 91 ; 14 Recall: 97 ; 4 f1-score: 94 ; 7 Support: 227 ; 23	Balance Accuracy: 50 Precision: 91 ; 10 Recall: 96 ; 4 f1-score: 93 ; 6 Support: 227 ; 23

Table 6.9: Relapses Binary with 50 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 50 Precision: 90 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 58 Precision: 92 ; 28 Recall: 94 ; 22 f1-score: 93 ; 24 Support: 227 ; 23
Gradient Boosting Classifier	Balance Accuracy: 51 Precision: 91 ; 25 Recall: 99 ; 4 f1-score: 95 ; 7 Support: 227 ; 23	Balance Accuracy: 52 Precision: 91 ; 14 Recall: 92 ; 13 f1-score: 91 ; 13 Support: 227 ; 23
KNN	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 57 Precision: 92 ; 17 Recall: 85 ; 30 f1-score: 89 ; 22 Support: 227 ; 23
Support vector machine	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 59 Precision: 93 ; 17 Recall: 81 ; 39 f1-score: 86 ; 24 Support: 227 ; 23
Naive Bayes	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23
Random Forest	Balance Accuracy: 52 Precision: 91 ; 100 Recall: 100 ; 4 f1-score: 95 ; 8 Support: 227 ; 23	Balance Accuracy: 55 Precision: 92 ; 18 Recall: 90 ; 22 f1-score: 91 ; 20 Support: 227 ; 23
XGboost	Balance Accuracy: 50 Precision: 91 ; 14 Recall: 97 ; 4 f1-score: 94 ; 7 Support: 227 ; 23	Balance Accuracy: 52 Precision: 91 ; 18 Recall: 96 ; 9 f1-score: 94 ; 12 Support: 227 ; 23

Table 6.10: Relapses Binary with 60 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 50 Precision: 90 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 59 Precision: 93 ; 16 Recall: 80 ; 39 f1-score: 86 ; 23 Support: 227 ; 23
Gradient Boosting Classifier	Balance Accuracy: 51 Precision: 91 ; 25 Recall: 99 ; 4 f1-score: 95 ; 7 Support: 227 ; 23	Balance Accuracy: 52 Precision: 91 ; 18 Recall: 96 ; 9 f1-score: 94 ; 12 Support: 227 ; 23
KNN	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 59 Precision: 93 ; 18 Recall: 84 ; 35 f1-score: 88 ; 24 Support: 227 ; 23
Support vector machine	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 59 Precision: 93 ; 16 Recall: 80 ; 39 f1-score: 86 ; 23 Support: 227 ; 23
Naive Bayes	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23
Random Forest	Balance Accuracy: 52 Precision: 91 ; 100 Recall: 100 ; 4 f1-score: 95 ; 8 Support: 227 ; 23	Balance Accuracy: 55 Precision: 92 ; 17 Recall: 89 ; 22 f1-score: 91 ; 19 Support: 227 ; 23
XGboost	Balance Accuracy: 50 Precision: 91 ; 14 Recall: 97 ; 4 f1-score: 94 ; 7 Support: 227 ; 23	Balance Accuracy: 51 Precision: 91 ; 13 Recall: 94 ; 9 f1-score: 93 ; 11 Support: 227 ; 23

Table 6.11: Relapses Binary with 65 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 50 Precision: 90 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 65 Precision: 94 ; 18 Recall: 74 ; 57 f1-score: 83 ; 28 Support: 227 ; 23
Gradient Boosting Classifier	Balance Accuracy: 51 Precision: 91 ; 25 Recall: 99 ; 4 f1-score: 95 ; 7 Support: 227 ; 23	Balance Accuracy: 52 Precision: 91 ; 17 Recall: 96 ; 9 f1-score: 93 ; 11 Support: 227 ; 23
KNN	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 56 Precision: 92 ; 17 Recall: 87 ; 26 f1-score: 90 ; 21 Support: 227 ; 23
Support vector machine	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 60 Precision: 93 ; 16 Recall: 77 ; 43 f1-score: 84 ; 24 Support: 227 ; 23
Naive Bayes	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23
Random Forest	Balance Accuracy: 52 Precision: 91 ; 100 Recall: 100 ; 4 f1-score: 95 ; 8 Support: 227 ; 23	Balance Accuracy: 53 Precision: 91 ; 14 Recall: 89 ; 17 f1-score: 90 ; 16 Support: 227 ; 23
XGboost	Balance Accuracy: 50 Precision: 91 ; 14 Recall: 97 ; 4 f1-score: 94 ; 7 Support: 227 ; 23	Balance Accuracy: 50 Precision: 91 ; 11 Recall: 93 ; 9 f1-score: 92 ; 10 Support: 227 ; 23

Table 6.12: Relapses Binary with 75 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 50 Precision: 90 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 63 Precision: 94 ; 16 Recall: 70 ; 57 f1-score: 81 ; 25 Support: 227 ; 23
Gradient Boosting Classifier	Balance Accuracy: 51 Precision: 91 ; 25 Recall: 99 ; 4 f1-score: 95 ; 7 Support: 227 ; 23	Balance Accuracy: 53 Precision: 91 ; 19 Recall: 94 ; 13 f1-score: 93 ; 15 Support: 227 ; 23
KNN	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 62 Precision: 93 ; 22 Recall: 86 ; 39 f1-score: 89 ; 28 Support: 227 ; 23
Support vector machine	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 57 Precision: 93 ; 14 Recall: 72 ; 43 f1-score: 81 ; 21 Support: 227 ; 23
Naive Bayes	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23
Random Forest	Balance Accuracy: 52 Precision: 91 ; 100 Recall: 100 ; 4 f1-score: 95 ; 8 Support: 227 ; 23	Balance Accuracy: 57 Precision: 92 ; 18 Recall: 88 ; 26 f1-score: 90 ; 21 Support: 227 ; 23
XGboost	Balance Accuracy: 50 Precision: 91 ; 14 Recall: 97 ; 4 f1-score: 94 ; 7 Support: 227 ; 23	Balance Accuracy: 51 Precision: 91 ; 14 Recall: 95 ; 9 f1-score: 93 ; 11 Support: 227 ; 23

Table 6.13: Relapses Binary with 85 percent re-balance class

Model Name	Imbalance dataset	Re-Balanced dataset
Logistic Regression	Balance Accuracy: 50 Precision: 90 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 66 Precision: 95 ; 17 Recall: 67 ; 65 f1-score: 79 ; 27 Support: 227 ; 23
Gradient Boosting Classifier	Balance Accuracy: 51 Precision: 91 ; 25 Recall: 99 ; 4 f1-score: 95 ; 7 Support: 227 ; 23	Balance Accuracy: 54 Precision: 92 ; 27 Recall: 96 ; 13 f1-score: 94 ; 18 Support: 227 ; 23
KNN	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 61 Precision: 93 ; 20 Recall: 84 ; 39 f1-score: 88 ; 26 Support: 227 ; 23
Support vector machine	Balance Accuracy: 50 Precision: 91 ; 0 Recall: 100 ; 0 f1-score: 95 ; 0 Support: 227 ; 23	Balance Accuracy: 55 Precision: 92 ; 12 Recall: 71 ; 39 f1-score: 80 ; 19 Support: 227 ; 23
Naive Bayes	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23	Balance Accuracy: 57 Precision: 98 ; 11 Recall: 19 ; 96 f1-score: 32 ; 19 Support: 227 ; 23
Random Forest	Balance Accuracy: 52 Precision: 91 ; 100 Recall: 100 ; 4 f1-score: 95 ; 8 Support: 227 ; 23	Balance Accuracy: 57 Precision: 92 ; 19 Recall: 89 ; 26 f1-score: 90 ; 22 Support: 227 ; 23
XGboost	Balance Accuracy: 50 Precision: 91 ; 14 Recall: 97 ; 4 f1-score: 94 ; 7 Support: 227 ; 23	Balance Accuracy: 50 Precision: 91 ; 10 Recall: 96 ; 4 f1-score: 93 ; 6 Support: 227 ; 23

Table 6.14: Relapses Binary with 100 percent re-balance class

Chapter 7

Conclusions

This thesis addressed the critical issue of class imbalance in medical predictive modeling, focusing specifically on binary outcomes for complications and relapses in post-operative hemorrhoid surgery patients. By employing advanced rebalancing techniques and hyperparameter optimization, we aimed to enhance the predictive accuracy and reliability of machine learning models for healthcare applications.

In this study, we systematically examined the impact of class imbalance on predictive modeling for two binary target variables: *Complication Binary* and *Relapses Binary*. Class imbalance is a significant challenge in medical machine learning, as the underrepresented class often holds vital clinical importance. To address this, we utilized the Synthetic Minority Oversampling Technique (SMOTE) to rebalance the datasets and systematically evaluated its impact on model performance. The influence of SMOTE oversampling ratios was assessed across a range from 40% to 100%, providing insights into their effects on model behavior. The evaluation employed key metrics such as balanced accuracy, precision, recall, F1-score, and support, offering a comprehensive understanding of model performance under different class distributions.

For the **Binary Complication** target, both **Gradient Boosting Classifier** and **Random Forest** demonstrated strong, consistent performance with high balanced accuracy and reasonable trade-offs between precision and recall. These models are particularly robust across various rebalancing ratios, making them excellent choices for handling imbalanced datasets like "Complication Binary.". These models' ability to maintain high performance with slight improvements across different rebalancing ratios suggests they are well-suited for predictive modeling in clinical settings where class imbalance is a significant challenge.

Similarly, for the **Binary Relapses** target, both **Gradient Boosting Classifier** and **Random Forest** demonstrated strong, consistent performance with improved balanced accuracy and reasonable trade-offs between precision and recall. These models' ability to maintain high performance with increasing rebalancing ratios suggests they are well-suited for predictive modeling in clinical settings where class imbalance is a significant challenge.

For both **Binary Complication** and **Binary Relapses**, **XGBoost** demonstrated consistent performance across different rebalancing ratios, with slight improvements in recall and balanced accuracy, making it suitable for predictive modeling in imbalanced datasets. This adaptability, combined with its scalability and speed, makes it a strong candidate for handling class imbalance effectively.

To further enhance model performance, we conducted hyperparameter optimization using grid

search. This systematic approach ensured that the machine learning models were tuned to their optimal configurations, enabling robust conclusions about their performance in handling imbalanced datasets.

Efforts to model multiclass versions of complications and relapses were met with substantial challenges due to extreme class imbalance and poor representation of minority classes. Despite attempts to consolidate and reduce the number of classes, performance remained unsatisfactory. This underscores the complexity of multiclass predictive tasks in highly imbalanced datasets and reaffirms the decision to focus on binary classification.

This study makes several important contributions to the field of predictive modeling in healthcare:

- Systematic evaluation of SMOTE oversampling ratios, providing actionable insights into how varying levels of oversampling influence predictive performance.
- Identification of optimal SMOTE ratios for different target variables: 40%-60% for *Binary Complication* and *Binary Relapses*, both with the Gradient Boosting Classifier.
- Integration of grid search for hyperparameter tuning, ensuring that models achieve their best configurations for imbalanced datasets.

While this thesis provides valuable insights, several limitations and opportunities for future research should be noted. The dataset was limited to a single hospital's patient population, which may affect the generalizability of the findings. Future studies could extend this work by incorporating larger and more diverse datasets to validate the models' applicability in varied clinical settings.

Additionally, exploring alternative rebalancing techniques, such as ADASYN or ensemble-based methods, could provide improved performance in handling class imbalance. Investigating novel strategies like cost-sensitive learning or hybrid approaches for multiclass predictions may address challenges posed by extreme imbalance in multiclass datasets. Further research could also integrate interpretability tools, such as SHAP (SHapley Additive exPlanations), to provide deeper insights into feature importance and model decision-making, thereby increasing trust in machine learning applications in healthcare.

In conclusion, this thesis has demonstrated the effectiveness of SMOTE in addressing class imbalance and improving the predictive performance of machine learning models for binary complications and relapses. By systematically optimizing SMOTE ratios and fine-tuning models through grid search, we achieved significant improvements in predictive accuracy and reliability. These findings contribute to the broader field of machine learning in healthcare, laying a foundation for future research and practical implementation in imbalanced datasets.

References

[1] Rabia Emhamed Al Mamlook, Lee J Wells, and Robert Sawyer. Machine-learning models for predicting surgical site infections using patient pre-operative risk and surgical procedure factors. *American journal of infection control*, 51(5):544–550, 2023.

[2] Abdullah Alanazi. Using machine learning for healthcare challenges and opportunities. *Informatics in Medicine Unlocked*, 30:100924, 2022.

[3] Saeed Ali Alsareii, Mohsin Raza, Abdulrahman Manaa Alamri, Mansour Yousef AlAsmari, Muhammad Irfan, Umar Khan, and Muhammad Awais. Machine learning and internet of things enabled monitoring of post-surgery patients: A pilot study. *Sensors*, 22(4):1420, 2022.

[4] Candice Bentéjac, Anna Csörgő, and Gonzalo Martínez-Muñoz. A comparative analysis of gradient boosting algorithms. *Artificial Intelligence Review*, 54:1937–1967, 2021.

[5] D.E. Berg. Surface roughness effects on a mach 6 turbulent boundary layer. *AIAA Journal*, 17(9):929–930, 1979.

[6] Azra Bihorac, Tezcan Ozrazgat-Baslanti, Ashkan Ebadi, Amir Motaei, Mohcine Madkour, Panagote M Pardalos, Gloria Lipori, William R Hogan, Philip A Efron, Frederick Moore, et al. Mysurgeryrisk: development and validation of a machine-learning risk algorithm for major complications and death after surgery. *Annals of surgery*, 269(4):652–662, 2019.

[7] Alison Callahan and Nigam H Shah. Machine learning in healthcare. In *Key advances in clinical informatics*, pages 279–291. Elsevier, 2017.

[8] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pages 785–794, 2016.

[9] Daniele Crocetti, F. Guerra, Livia De Anna, F. Velluti, A. Giuliani, F. La Torre, et al. Surgical treatment of postoperative complications following procedure for prolapsed hemorrhoids (pph). *Techniques in Coloproctology*, 13:349–349, 2009.

[10] Catarina Alexandra Teixeira Ribeiro Azevedo Da. *Challenges and Pitfalls in the Treatment of Sepsis: Knowledge of Pathophysiology Towards Better Diagnosis*. PhD thesis, Universidade de Lisboa, 2022.

[11] Vishal Dutt, Pramod Singh Rathore, and Kapil Chauhan. Support vector in healthcare using svm/pso in various domains: A review. *Swarm Intelligence Optimization: Algorithms and Applications*, pages 291–307, 2020.

[12] A. Famili, Wei-Min Shen, Richard Weber, and Evangelos Simoudis. Data preprocessing and intelligent data analysis. *Intelligent Data Analysis*, 1(1):3–23, 1997.

[13] Nadia Fathallah, Amine Alam, Anne L Rentien, Giorgio La Greca, Joshua Co, Elise Pommaret, Amélie Barré, Stéphane Kegne, Hélène Beaussier, Lucas Spindler, et al. Hemorrhoidal disease: Epidemiological study and analysis of predictive factors for surgical management. *Journal of Visceral Surgery*, 2024.

[14] Hafsa Habehh and Suril Gohel. Machine learning in healthcare. *Current genomics*, 22(4):291, 2021.

[15] Jonathan E Helm, Adel Alaeddini, Jon M Stauffer, Kurt M Bretthauer, and Ted A Skolarus. Reducing hospital readmissions by integrating empirical prediction with resource optimization. *Production and Operations Management*, 25(2):233–257, 2016.

[16] Mohammad Hossin and Md Nasir Sulaiman. A review on evaluation metrics for data classification evaluations. *International journal of data mining & knowledge management process*, 5 (2):1, 2015.

[17] Shujun Huang, Nianguang Cai, Pedro Penzuti Pacheco, Shavira Narrandes, Yang Wang, and Wayne Xu. Applications of support vector machine (svm) learning in cancer genomics. *Cancer genomics & proteomics*, 15(1):41–51, 2018.

[18] GP Joshi and EAM Neugebauer. Evidence-based management of pain after haemorrhoidectomy surgery. *Journal of British Surgery*, 97(8):1155–1168, 2010.

[19] Harsurinder Kaur, Husanbir Singh Pannu, and Avleen Kaur Malhi. A systematic review on imbalanced data challenges in machine learning: Applications and solutions. *ACM computing surveys (CSUR)*, 52(4):1–36, 2019.

[20] Michael P LaValley. Logistic regression. *Circulation*, 117(18):2395–2399, 2008.

[21] Christoph Mallmann, Mike Ralf Langenbach, Razvan-Valentin Florescu, Andreas Köhler, Jörg Barkus, Jörg-Peter Ritz, Florian Gebauer, Rolf Lefering, and Lars Boenicke. Parameters predicting postoperative pain and quality of life after hemorrhoidectomy: follow-up results from a prospective multicenter randomized trial. *International Journal of Colorectal Disease*, 38(1): 262, 2023.

[22] Alexander Meyer, Dina Zverinski, Boris Pfahringer, Jörg Kempfert, Titus Kuehne, Simon H Sündermann, Christof Stamm, Thomas Hofmann, Volkmar Falk, and Carsten Eickhoff. Machine learning for real-time prediction of complications in critical care: a retrospective study. *The Lancet Respiratory Medicine*, 6(12):905–914, 2018.

[23] Geeta Sandeep Nadella, Snehal Satish, Karthik Meduri, and Sai Sravan Meduri. A systematic literature review of advancements, challenges and future directions of ai and ml in healthcare. *International Journal of Machine Learning for Sustainable Development*, 5(3):115–130, 2023.

[24] David S Palmer, Noel M O'Boyle, Robert C Glen, and John BO Mitchell. Random forest models to predict aqueous solubility. *Journal of chemical information and modeling*, 47(1): 150–158, 2007.

[25] Marcello Picchio, Ettore Greco, Annalisa Di Filippo, Giuseppe Marino, Francesco Stipa, and Erasmo Spaziani. Clinical outcome following hemorrhoid surgery: a narrative review. *Indian Journal of Surgery*, 77:1301–1307, 2015.

[26] Hema Sekhar Reddy Rajula, Giuseppe Verlato, Mirko Manchia, Nadia Antonucci, and Vassilios Fanos. Comparison of conventional statistical methods with machine learning in medicine: diagnosis, drug development, and treatment. *Medicina*, 56(9):455, 2020.

[27] Iestyn M Shapey and Mustafa Sultan. Machine learning for prediction of postoperative complications after hepato-biliary and pancreatic surgery. *Artificial Intelligence Surgery*, 3(1):1–13, 2023.

[28] Swati Shilaskar, Ashok Ghatol, and Prashant Chatur. Medical decision support system for extremely imbalanced datasets. *Information Sciences*, 384:205–219, 2017.

[29] S Sreejith, H Khanna Nehemiah, and Arputharaj Kannan. Clinical data classification using an enhanced smote and chaotic evolutionary feature selection. *Computers in Biology and Medicine*, 126:103991, 2020.

[30] Gregor Stiglic, Primoz Kocbek, Nino Fijacko, Marinka Zitnik, Katrien Verbert, and Leona Cilar. Interpretability of machine learning-based prediction models in healthcare. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 10(5):e1379, 2020.

[31] Richard PG ten Broek, N Kok-Krant, EA Bakkum, RP Bleichrodt, and H Van Goor. Different surgical techniques to reduce post-operative adhesion formation: a systematic review and meta-analysis. *Human reproduction update*, 19(1):12–25, 2013.

[32] Chunming Xu and Scott A Jackson. Machine learning and complex biological data, 2019.

[33] Harry Zhang. The optimality of naive bayes. *Aa*, 1(2):3, 2004.

[34] Yulong Zhang, Shiwei Sun, and Zhenguo Han. Establishment and validation of clinical prediction model for hemorrhoid recurrence after procedure for prolapse and hemorrhoids. *Medicine*, 102(26):e34062, 2023.

[35] Zhongheng Zhang. Introduction to machine learning: k-nearest neighbors. *Annals of translational medicine*, 4(11), 2016.

Acknowledgements

I would like to express my deepest gratitude to my supervisors, Dr. Andrea Manno and Prof. Fabrizio Rossi, for their invaluable guidance, patience, and continuous support throughout this journey. Their expertise and encouragement have been instrumental in the completion of this thesis, and I am truly grateful for their mentorship.

I also want to extend my heartfelt thanks to my parents, family, and friends. Their support and belief in me have been a constant source of strength, and I could not have accomplished this without them.

Thank you all for being there every step of the way.

Abdoulie J Bah, L'Aquila, December 2024.